

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of MSc Biotechnology/Biochemistry Examination March 2018

Course code & Name: MS713 Cell Biology

Date: 28/03/2018 Day: Wednesday Time: 10:00 a.m To 01:00 p.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions.

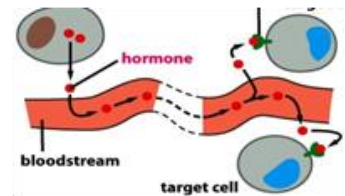
20

- Which of the following is present in plant cells?
(i) Cell wall (ii) Vacuole
(iii) Chloroplast (iv) All of the above
- Which of the following components are not present in lipid rafts?
(i) Glycoproteins (ii) Cholesterol
(iii) Proteoglycans (iv) None of the above
- Gramicidin A is an example of _____.
(i) Ionophores (ii) Ion channels
(iii) Cell signaling molecule (iv) Uniport transporter
- Electron Spin Resonance technique can be used to study
(i) Flip-flop movements of lipids (ii) Lateral movement of lipids
(iii) Lateral movement of proteins (iv) All of the above

5. O-linked glycosylation occurs in _____
- (i) lysosomes (ii) endoplasmic reticulum
(iii) peroxisomes (iv) Golgi apparatus
6. Clathrin coated vesicles are involved in _____.
(i) transcytosis (ii) protein synthesis
(iii) vesicular transport of proteins (iv) nuclear transport of proteins
7. Which of the following does NOT occur within the nucleus?
(i) Histone protein import (ii) Ribosomal subunit assembly
(iii) rRNA synthesis (iv) None of the above

8. Identify the form of signaling.

- (i) Endocrine
(ii) Synaptic
(iii) Juxtacrine
(iv) Paracrine



9. Which of the following represents secondary active transport mechanism?
(i) Na^+ symport of glucose (ii) HCl transport in parietal cells
(iii) Na^+ K^+ ATPase (iv) ABC transporters
10. Calmodulin acts as the universal intracellular receptor for _____.
(i) IP_3 (ii) Ca^{2+}
(iii) cAMP (iv) Cytokines
11. Which of the following substitutes the functions of lysosomes in plant cells?
(i) Vacuoles (ii) Early endosomes
(iii) Amyloplast (iv) Glyoxysomes
12. Tight junctions are site for connection of _____.
(i) actin filaments (ii) microtubules
(iii) intermediate filaments (iv) integrins
13. Amyloplasts are a type of _____.
(i) ECM component (ii) proteoglycans
(iii) vesicles (iv) Plastids

14. Which of the following is essential for formation of gap junctions?
- (i) Claudins
 - (ii) Clathrin
 - (iii) Integrins
 - (iv) Connexins
15. Caspase-3 activation is a hallmark of _____.
- (i) APC activation
 - (ii) DNA replication
 - (iii) G₀ phase
 - (iv) Apoptosis
16. DNA replication takes place during
- (i) S – phase
 - (ii) G₀ phase
 - (iii) G₁ phase
 - (iv) Prophase
17. Which of the following ECM protein is highly hydroxylated?
- (i) Elastin
 - (ii) Collagen
 - (iii) Fimbrin
 - (iv) Fibronectin
18. Ribosomes are associated with _____.
- (i) Smooth Endoplasmic reticulum
 - (ii) Golgi apparatus
 - (iii) Rough Endoplasmic reticulum
 - (iv) Nuclear envelope
19. Na⁺-K⁺ ATPase is best described as _____.
- (i) Uniporter
 - (ii) Antiporter
 - (iii) Symporter
 - (iv) Passive transporter
20. Restriction point is the time in cell cycle when cell becomes committed to enter from _____phase to _____phase.
- (i) G₁ to G₂
 - (ii) G₀ to G₁
 - (iii) G₁ to S
 - (iv) G₂ to M

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Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I

Q - II Answer the following questions as directed 20

- (A) Discuss Fluid Mosaic Model of membrane structure. Explain various types of membrane proteins 04**

OR

- (A) Describe structure of a typical neuronal cell and explain the mechanism of action potential 04**

(B) Write a short note on the following (Any TWO) 06

- (i) Fluorescence recovery after photobleaching (FRAP) technique
- (ii) Structure and organization of lipid rafts
- (iii) Active and Passive transport mechanisms across lipid bilayer

- (C) List down major phases of cell cycle. Discuss the stages of mitosis in detail. 04**

OR

- (C) Explain DNA damage check points 04**

(D) Write a short note on the following (Any TWO) 06

- (i) Apoptosis and its features
- (ii) M-Cdk activity and its regulation during the Cell Cycle
- (iii) Use of Yeast to discover mechanism of cell cycle regulation

SECTION - II

Q-III Answer the following questions as directed 30

(A) Explain the following (Any SIX) 18

- (i) Types of cell signaling receptors
- (ii) General mechanism of inter-cellular signaling
- (iii) NO signaling mechanism
- (iv) Components of ECM
- (v) Anchoring junctions
- (vi) Types of plastids
- (vii) Structure and function of Rough ER and Smooth ER
- (viii) Organization of nuclear envelope

(B) Answer the following (Any THREE) 12

- (i) Mechanism of insulin signaling through tyrosine kinase receptors
- (ii) Explain the mechanism of activation of G-protein coupled receptors.
Describe GPCR mediated signaling mechanism mediated by IP3
- (iii) Discuss in detail types of cytoskeletal filaments
- (iv) Mechanism of N-linked glycosylation of proteins
- (v) Discuss the typical structure of collagen fibril. Explain in detail the mechanism of collagen fibril assembly.